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Functional Near-Infrared Spectroscopy and Machine Learning For Neurocognitive and Mental-Health Screening

Literature Review

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1. Literature Review

In recent years, functional near-infrared spectroscopy (fNIRS) has become an increasingly valuable neuroimaging technique for the investigation of brain functionality and the dynamics of neurocognitive processes [4, 5]. One of the advantages of fNIRS is its portability and suitability for studies in realistic environments that cannot accommodate bulky scanner setups. In neurocognitive and mental-health screening contexts, however, fNIRS should be considered cautiously. Continuous-wave fNIRS does not estimate brain activity directly and does not provide a diagnostic biomarker by itself. Instead, it provides an indirect estimate of relative haemodynamic responses based on changes in oxyhaemoglobin and deoxyhaemoglobin levels [6, 7, 8].

Therefore, the scientifically defensible question is not whether fNIRS and machine-learning algorithms can directly diagnose mental-health conditions or neurodevelopmental disorders. A more accurate question is whether relative changes in frontal haemodynamic patterns can be preprocessed, represented, and classified under valid methodological conditions. This view aligns with current best practices and recent reproducibility guidance, which show that fNIRS outcomes are strongly influenced by signal quality, short-channel correction, data structuring, terminology, and evaluation strategy [6, 9, 10, 11, 12].

1.1. Scope of This Literature Analysis

The key challenge is not only obtaining high accuracy with a classifier. The more important question is whether the signal entered into the classifier is physiologically plausible. Current fNIRS methodology stresses that systemic physiology, superficial blood flow, motion, optode contact, preprocessing choices, and participant-wise evaluation must be controlled before a classification result can be interpreted [6, 7, 13, 9].

Several papers cited in this review contain examples of fNIRS classification and deep learning [1, 2, 14, 3, 15, 16]. Their relevance is conditional on the limitations of fNIRS data. The thesis therefore relies most strongly on methodological evidence concerning short-channel regression, physiological confounds, benchmarking, reproducible data organization, and data leakage [17, 18, 13, 19, 11, 12, 20].

1.2. fNIRS Signal Composition and Haemodynamic Estimates

An fNIRS signal is obtained by detecting near-infrared light intensity after scattering and absorption in biological tissue. Optical density (OD) can be calculated by subtracting the logarithm of the ratio between recorded and reference intensities:

$$\mathrm{OD}(\lambda,t) = -\log\left(\frac{I(\lambda,t)}{I_0(\lambda)}\right). \tag{1}$$

The corresponding change in optical density is defined as:

$$\Delta \mathrm{OD}(\lambda, t) = \mathrm{OD}(\lambda, t) - \mathrm{OD}(\lambda, t_0). \tag{2}$$

For continuous-wave fNIRS, optical-density signals are transformed into relative haemodynamic signal changes by applying the modified Beer-Lambert law. For a given wavelength λ, this relation can be written as:

$$\Delta A(\lambda) = \mathrm{DPF} \cdot r \cdot \left[\varepsilon_{\mathrm{HbO}}(\lambda)\Delta\mathrm{HbO} + \varepsilon_{\mathrm{HbR}}(\lambda)\Delta\mathrm{HbR}\right], \tag{3}$$

where DPF is the differential pathlength factor, r is the source-detector distance, $\varepsilon_{\mathrm{HbO}}$ and $\varepsilon_{\mathrm{HbR}}$ are the extinction coefficients of oxyhaemoglobin and deoxyhaemoglobin, and ΔHbO and ΔHbR are relative concentration changes [6, 8, 10]. For two wavelengths, the same relation can be represented as:

$$\begin{bmatrix} \Delta A(\lambda_1) \\ \Delta A(\lambda_2) \end{bmatrix} = \mathrm{DPF} \cdot r \begin{bmatrix} \varepsilon_{\mathrm{HbO}}(\lambda_1) & \varepsilon_{\mathrm{HbR}}(\lambda_1) \\ \varepsilon_{\mathrm{HbO}}(\lambda_2) & \varepsilon_{\mathrm{HbR}}(\lambda_2) \end{bmatrix} \begin{bmatrix} \Delta\mathrm{HbO} \\ \Delta\mathrm{HbR} \end{bmatrix}. \tag{4}$$

The fNIRS features used in this thesis are based on the following relative haemodynamic estimates:

$$\mathrm{HbDiff} = \Delta\mathrm{HbO} - \Delta\mathrm{HbR}, \tag{5}$$

$$\mathrm{HbT} = \Delta\mathrm{HbO} + \Delta\mathrm{HbR}. \tag{6}$$

These quantities should not be confused with oxygen saturation, because saturation depends on an estimation algorithm and hardware-specific assumptions [8, 10]. For example, a saturation quantity could be calculated as:

$$\mathrm{rSO_2 \text{ or } TSI} = \frac{\mathrm{HbO_2}}{\mathrm{HbO_2} + \mathrm{Hb}} \times 100, \tag{7}$$

but this term should be used only when it is actually calculated by the device or analysis software. Thus, the present research uses ΔHbO, ΔHbR, HbDiff, and HbT as relative haemodynamic features.

1.3. Methodological Evidence From fNIRS Studies

1.3.1. Physiological Confounds and Short-Channel Regression

One disadvantage of fNIRS is that long-distance channels receive mixed signals from cortical haemodynamics and superficial tissues, such as the scalp and skin. Superficial physiology includes systemic activity, blood-pressure changes, breathing, skin and scalp blood flow, motion artefacts, Mayer waves, and optode-contact changes [7, 13]. If these variables are not controlled, they can become the main

drivers of machine-learning model output.

One way to address this limitation is to use short-distance channels, because short channels are more sensitive to superficial physiological noise than long channels [17, 18, 19]. The following equation illustrates how a short-channel signal can serve as a nuisance variable for correcting a long-channel signal:

$$y_{\text{clean}} = y_{\text{long}} - \alpha y_{\text{short}}, \qquad \alpha = \frac{\langle y_{\text{short}}, y_{\text{long}} \rangle}{\langle y_{\text{short}}, y_{\text{short}} \rangle}. \tag{8}$$

Here, y_{long} is a measured long-distance signal, y_{short} is a short-channel signal, and y_{clean} is the corrected long-distance signal. The coefficient α defines the proportion of superficial haemodynamic activity shared between short and long channels.

More generally, this correction can be expressed using a general linear model:

$$Y = X_{\text{task}}\beta_{\text{task}} + X_{\text{SDC}}\beta_{\text{SDC}} + X_{\text{phys}}\beta_{\text{phys}} + \varepsilon, \tag{9}$$

where X_{task} includes task-related regressors, X_{SDC} includes short-distance-channel regressors, X_{phys} includes additional physiological nuisance variables where available, and ε includes the residual signal. This decomposition is especially important in mental-health applications, where intergroup task-related differences can be subtle and easily confounded by systemic physiology [7, 9, 13].

Recent reproducibility studies confirm the need to treat noise and preprocessing as part of scientific validity rather than as secondary details. The FRESH initiative showed that fNIRS findings are sensitive to signal quality, preprocessing steps, and researcher experience [9]. For this thesis, short-channel regression, artefact handling, and explicit reporting of preprocessing are therefore key steps before machine-learning techniques are applied.

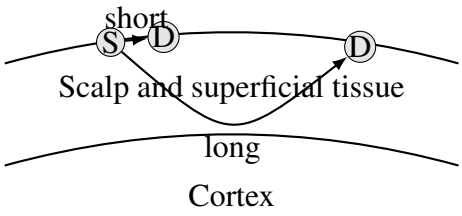


Fig. 1. Conceptual short- and long-separation fNIRS channels. Short channels estimate superficial physiology; long channels contain both superficial and cortical components.

1.3.2. Signal Structure and Representation for Machine Learning

One of the most important steps in machine learning with fNIRS data is signal representation. Unlike images, raw fNIRS data are not naturally represented in two-dimensional spatial coordinates. They are multichannel time series. For an analysis window, the input can be defined as:

$$X \in \mathrm{R}^{C \times T \times K}, \tag{10}$$

where C is the number of fNIRS channels, T is the number of time samples per channel, and K is the number of haemodynamic signals, such as ΔHbO, ΔHbR, HbDiff, and HbT. For N analysis windows, the full dataset can be written as:

$$\mathscr{X} \in \mathrm{R}^{N \times C \times T \times K}. \tag{11}$$

This representation is preferable to aggressive reshaping into a generic image format. Modern fNIRS deep-learning models show that CNN-LSTM, 1D convolutional, attentional, and temporal convolutional models can process temporal and spatial aspects of fNIRS signals without discarding their channel-by-time structure [2, 16, 14]. Classical machine-learning models can also use extracted features from these windows, provided that feature extraction and model evaluation are leakage-safe [11].

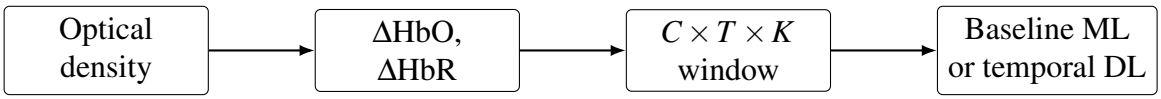


Fig. 2. Sequence-preserving fNIRS representation used for model input.

1.3.3. GASF-Based Task-State Classification

Some studies use image-based transformations of fNIRS data, such as Gramian Angular Summation Field (GASF). Wickramaratne and Mahmud used fNIRS signals transformed with GASF for three-class task classification, including mental arithmetic, motor imagery, and idle state [1]. This confirms that fNIRS data can contain task-related haemodynamic activity, although it also raises questions about whether GASF preserves physiologically meaningful structure.

The GASF representation is created by first normalizing the signal:

$$\tilde{x}_i = \frac{x_i - \min(X)}{\max(X) - \min(X)}. \tag{12}$$

Each sample is then converted into a polar-angle representation:

$$\varphi_i = \arccos(\tilde{x}_i), \tag{13}$$

and the matrix is constructed as:

$$\mathrm{GASF}_{ij} = \cos(\varphi_i + \varphi_j), \tag{14}$$

$$\mathrm{GASF}_{ij} = \tilde{x}_i \tilde{x}_j - \sqrt{1-\tilde{x}_i^2}\sqrt{1-\tilde{x}_j^2}. \tag{15}$$

Although GASF may be useful for designing image-based deep-learning models, this technique can distort the original physiological structure of fNIRS data. In this thesis, GASF is therefore treated as one possible representation from the literature rather than as the preferred approach. To retain physiological meaning, the window length, sampling frequency, channel distribution, and haemodynamic signal structure should remain explicit [2, 16, 11].

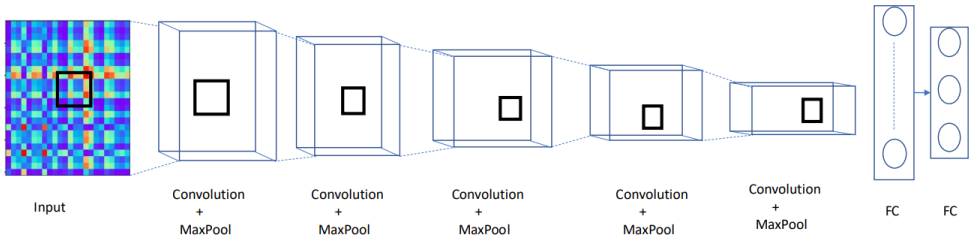


Fig. 3. Example convolutional neural network used for GASF-based fNIRS task classification [1].

1.3.4. Classical Machine Learning and Deep Learning Models

Classical machine-learning techniques remain important because they provide interpretable and reproducible baselines for deep-learning models. Common classical methods in fNIRS studies include linear discriminant analysis, logistic regression, support vector machines with linear or kernel methods, decision trees, random forests, and k-nearest neighbours [11, 14]. These models are usually applied to manually extracted features such as mean, slope, peak value, time-to-peak, variance, area under the curve, or spectral features. A feature extraction operation can be written as:

$$z_i = g(X_i), \tag{16}$$

where X_i is a haemodynamic window and z_i is a feature vector. The classifier then maps the feature vector into class probabilities:

$$\hat{y}_i = f_\theta(z_i). \tag{17}$$

Deep-learning models allow more flexible feature extraction without manual feature design. However, this increased flexibility also increases the risk of overfitting and leakage. Two common fNIRS deep-learning architectures use convolutional layers and recurrent layers, such as LSTM layers. CNN layers extract spatial or local temporal structures, whereas LSTM layers model temporal dependencies. One-dimensional convolution over time can be written as:

$$Y[i] = f\left(\sum_{j=1}^{k} X[i+j-1]\cdot W[j] + b\right), \tag{18}$$

where k is the convolution-kernel size, W is the kernel, b is the bias, and $f(\cdot)$ is a nonlinear activation function. A commonly used activation function is the rectified linear unit:

$$\mathrm{ReLU}(x) = \max(0, x). \tag{19}$$

Pooling can be applied to reduce the temporal dimension:

$$Y[i] = \max(X[i\cdot p : (i+1)\cdot p]). \tag{20}$$

CNN-LSTM models are relevant because fNIRS responses evolve over time. Recent research also explores spatiotemporal convolution, temporal convolutional networks, and attention mechanisms [14, 16]. In general, the final classifier of such a model can be expressed as:

$$\hat{y} = f_\theta(X), \quad X \in \mathrm{R}^{C\times T\times K}. \tag{21}$$

The loss function commonly used for classification problems is cross-entropy:

$$\mathscr{L} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{m=1}^{M} y_{im}\log(\hat{p}_{im}), \tag{22}$$

where M is the number of classes, y_{im} is the class indicator, and $\hat{p}_{im}$ is the predicted probability for class m.

1.3.5. CNN-LSTM Neural Network Model for fNIRS Workload Classification

Convolutional and LSTM layers are potentially useful for building an architecture capable of processing two forms of variation in fNIRS data. CNN layers learn local patterns, whereas LSTM layers model dependencies between samples [2]. Nevertheless, high reported accuracy in workload classification cannot imply diagnostic ability, because workload classes are experimentally defined and may reflect many non-clinical factors.

1.3.6. Task-State, Workload, and Walking-Task Classification

Modern literature supports fNIRS machine learning for task-state, workload, and walking-task classification more strongly than for standalone diagnosis. Khan et al. show the benefits of deep learning for cognitive workload modelling [2]. This is relevant because workload is related to mental effort. Nevertheless, workload classification cannot be equated with mental-health screening: a classification

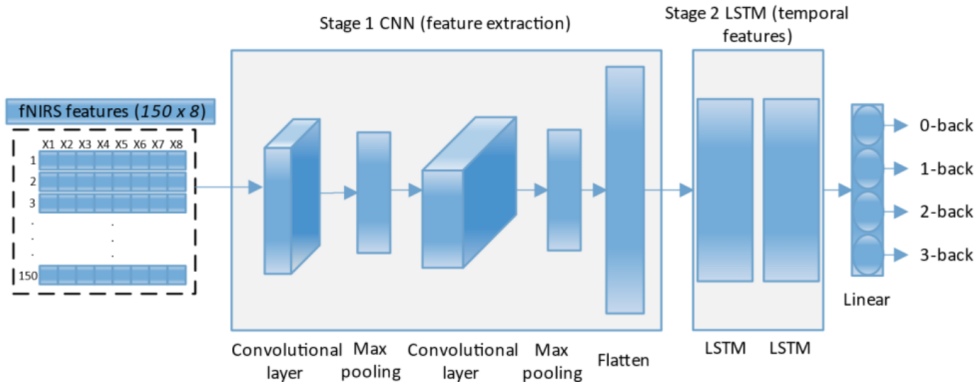


Fig. 4. Example CNN-LSTM architecture for fNIRS workload classification [2].

task is defined experimentally, whereas a diagnosis is defined by clinical criteria.

Ma et al. developed a deep-learning model to distinguish walking tasks in older adults [3]. This work demonstrates the potential of fNIRS and deep learning for classification in motor-cognitive paradigms. At the same time, motion artefacts become especially important during walking, which increases the need for careful preprocessing and leakage-free evaluation [3, 7, 12].

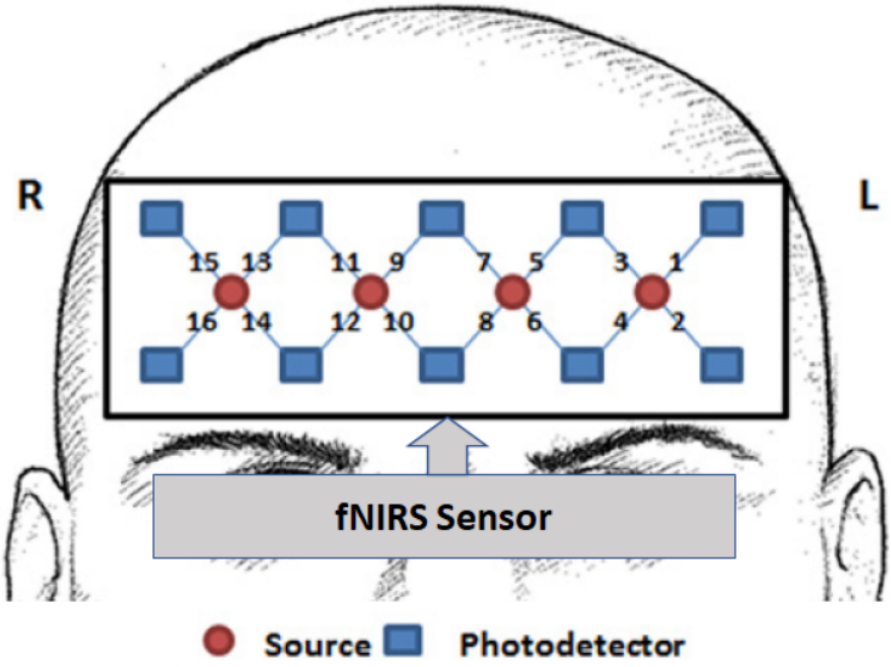


Fig. 5. Example fNIRS sensor placement used in walking-task classification [3].

1.3.7. Cognitive-Effort Modelling Framework

A cognitive-effort modelling framework is relevant because it allows the analysis to consider individual effort in addition to task performance [15]. Cognitive effort must be defined independently from performance. Standardization can be written as:

$$z = \frac{x - \mu}{\sigma}. \tag{23}$$

For performance and cognitive-load variables, this becomes:

$$P_z(s,i) = \frac{P(s,i) - \mu(P_i)}{\sigma(P_i)}, \tag{24}$$

$$M_z(s,i) = \frac{M(s,i) - \mu(M_i)}{\sigma(M_i)}. \tag{25}$$

Relative neural efficiency (RNE) and relative neural involvement (RNI) can then be written as:

$$\mathrm{RNE} = \frac{P_z - M_z}{\sqrt{2}}, \tag{26}$$

$$\mathrm{RNI} = \frac{P_z + M_z}{\sqrt{2}}. \tag{27}$$

RNE and RNI may help distinguish participants in terms of cognitive effort, but they cannot serve as reliable indicators of clinical disorders. They are therefore useful only as possible neurocognitive task-classification variables.

1.3.8. Leakage-Free Evaluation of Models

Several machine-learning issues create major challenges for fNIRS research, especially leakage and overfitting. When time windows from the same participant are divided between training, validation, and test sets, a classifier may learn participant-specific features, superficial physiology, motion patterns, or optode-placement differences instead of task-related haemodynamic responses [11, 12].

A leakage-free procedure can be described by splitting subjects first:

$$S = S_{\text{train}} \cup S_{\text{val}} \cup S_{\text{test}}, \tag{28}$$

where:

$$S_{\text{train}} \cap S_{\text{val}} = \emptyset, \quad S_{\text{train}} \cap S_{\text{test}} = \emptyset, \quad S_{\text{val}} \cap S_{\text{test}} = \emptyset. \tag{29}$$

If the time windows from participant i are denoted as $w_{i,t}$, then:

$$i \in S_k \Rightarrow w_{i,t} \in D_k \quad \forall t. \tag{30}$$

This means that windows from a participant can belong to the training set, validation set, or test set, but cannot be divided across those sets. Splitting should occur before scaling, feature selection, dimensionality reduction, hyperparameter tuning, or model selection [11].

BenchNIRS provides a useful background for leakage-safe fNIRS machine learning because it evaluates several models using public fNIRS Hub datasets and nested cross-validation [11]. It also shows that performance on unseen data is lower than many optimistic reports suggest. Nested cross-validation can be expressed as:

$$\theta^* = \arg\min_{\theta} \frac{1}{K} \sum_{k=1}^{K} \mathscr{L}_{\text{val}}^{(k)}(\theta), \tag{31}$$

where θ^* denotes hyperparameters selected in the inner loop. The selected model is then tested on the outer set of participants:

$$\text{Score}_{\text{test}} = \text{Metric}(f_{\theta^*}(X_{\text{test}}), y_{\text{test}}). \tag{32}$$

If classes are unbalanced, balanced accuracy is more informative than ordinary accuracy:

$$\text{Balanced Accuracy} = \frac{1}{M} \sum_{m=1}^{M} \frac{\text{TP}_m}{\text{TP}_m + \text{FN}_m}, \tag{33}$$

where M is the number of classes, TP_m is the number of true positives for class m, and FN_m is the number of false negatives. F1-score, sensitivity, specificity, and confusion matrices should also be reported.

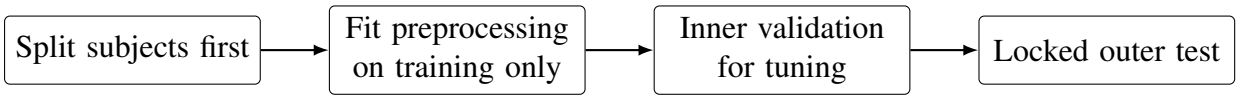


Fig. 6. Leakage-safe evaluation pipeline for fNIRS machine learning.

1.3.9. Reproducibility Guidelines, Standardisation, and Wearable fNIRS

The diversity of fNIRS preprocessing and modelling approaches makes reproducibility a central concern. FRESH shows that findings can vary with signal quality, analysis pipelines, and researcher expertise [9]. Consequently, machine-learning models should not be compared unless preprocessing and data quality are reported clearly.

NIRS-BIDS is important for this thesis because it extends the Brain Imaging Data Structure to near-infrared spectroscopy [20]. It supports standardisation of subject folders, tasks, events, channels, optodes, coordinate systems, and SNIRF-compatible data. The broader BIDS framework also provides a model for reproducible neuroimaging data organization [21]. Organizing fNIRS data according to a BIDS-compatible or BIDS-inspired scheme is therefore an important future step for the thesis.

Wearable fNIRS is another reason why this research direction may become more relevant in the future. Dense-sampling wearable platforms may support repeated measurements in natural environments and precision functional neuroimaging [22]. However, current evidence remains preliminary. The safest conclusion is that wearable fNIRS may enable longitudinal studies and individual functional mapping, but more diverse clinical cohorts are required before claims about disorder screening can be made.

1.3.10. Connectivity-Aware and Graph-Based Extensions

Most fNIRS classification studies treat channels as independent variables. Recent work has started to model fNIRS data as networks, where nodes represent channels or cortical areas and edges represent relationships between those areas [23]. This may be relevant because brain functions are not isolated to single cortical locations.

A graph convolutional layer can be represented as:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right), \tag{34}$$

where $\tilde{A}$ is an adjacency matrix with self-edges, $\tilde{D}$ is a degree matrix, $H^{(l)}$ is the feature matrix at layer l, $W^{(l)}$ is a trainable weight matrix, and σ is a nonlinear activation function. Tang et al. predicted cortical-thalamic functional connectivity from fNIRS-derived functional data using graph convolutional networks [23]. This approach does not remove the depth limitation of fNIRS, but it shows that graph-based methods can model connectivity-aware information. In this thesis, graph-based approaches should therefore be considered future additions rather than the immediate methodological step.

1.3.11. Cross-Study Interpretation

The literature provides a cautious but sound basis for the thesis position. fNIRS combined with machine-learning techniques can be used to classify task patterns, cognitive workload, and cognitive effort in walking tasks [1, 2, 3, 15, 16, 11]. At the same time, fNIRS and machine learning are not sufficient for conclusive diagnosis of mental disorders or other brain conditions. This caution is supported by broader neuroimaging evidence showing that clinical classification can remain weak even in large multi-site datasets and that human neurobiology does not map cleanly onto simple diagnostic labels [24, 25].

Therefore, the validity of fNIRS machine learning should be tested by evaluating the entire pipeline:

$$\begin{aligned}&\text{Raw intensity} \rightarrow \text{Optical density} \rightarrow \Delta\text{HbO}, \Delta\text{HbR} \rightarrow \text{Short-channel correction}\\ &\rightarrow \text{Windowed representation} \rightarrow \text{Leakage-safe model training} \rightarrow \text{Unseen-participant evaluation.}\end{aligned} \tag{35}$$

If one step in this chain is problematic, classification may produce apparently accurate results without

physiological meaning. For this reason, the thesis should evaluate classifier performance by generalisation to unseen subjects, stability across folds, comparison with baseline methods, and interpretability of channel and haemodynamic contributions.

1.3.12. Methodological Implications

Traditional machine-learning models are important baselines. At minimum, logistic regression or linear discriminant analysis, linear support vector machine, RBF-kernel support vector machine, and random forest should be tested. CNN, LSTM, CNN-LSTM, and temporal-convolutional models become useful only if they generalize better than these baselines [11, 2, 16].

The reviewed literature supports further development of this research only under careful methodological conditions. The thesis should avoid claiming that fNIRS diagnoses mental-health or neurodevelopmental conditions. A stronger and more defensible claim is that task-evoked fNIRS signals, analysed with short-channel correction and participant-wise validation, can reveal relative haemodynamic patterns associated with selected neurocognitive task conditions.

2. Methodology

The thesis topic sits in an attractive but risky research space. fNIRS offers a portable and non-invasive way to observe relative haemodynamic activity during cognitive, motor, educational, or social tasks. However, mental-health and neurodevelopmental conditions rarely map to one clean physiological marker. A model may find statistical structure in the signal without learning a clinically meaningful pattern.

For that reason, the defensible framing is not that fNIRS diagnoses mental disease. The stronger position is that task-evoked fNIRS, combined with leakage-safe machine learning, may contribute to objective assessment by modelling relative neurocognitive response patterns.

2.1. Methodological Risks

The current methodology must control four risks. First, extracerebral systemic physiology can contaminate long-separation fNIRS channels, so short-distance channels should be used as nuisance regressors where available [7, 17, 18, 19]. Second, fNIRS should be represented as channel-by-time haemodynamic data rather than arbitrarily resized into a generic image grid [2, 16]. Third, HbDiff and HbT are relative haemodynamic quantities and should not be described as tissue oxygen saturation [6, 8]. Fourth, model evaluation must avoid leakage from participant identity, preprocessing, hyperparameter tuning, and overlapping windows [11, 12].

2.2. Thesis Positioning

The reviewed literature supports task-state, workload, walking-task, and cognitive-effort modelling more strongly than stand-alone clinical diagnosis [1, 14, 3, 15]. A strong and defensible aim is to investigate whether task-evoked fNIRS signals, analysed with short-channel correction and subject-wise validation, can reveal objective relative haemodynamic patterns associated with selected neurocognitive indicators.

3. Conclusions

This literature analysis shows that the thesis methodology can be supported by a modest, but solid claim: the combination of fNIRS and machine learning is currently better supported for classifying relative haemodynamic task patterns than for clinical diagnosis.

To that end, the updated methodology considers the following factors: short-distance channel regression for eliminating extracerebral effects, preservation of channel-by-time representation, distinction of relative haemodynamic patterns, and reporting of oxygenation only when explicitly estimated by the instrument and algorithm.

The future research should address leakage problems, employ nested hyperparameter optimization, perform preprocessing in a leakage-safe manner, use robust baselines, and implement model interpretability techniques.

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